

(a) The polymerase chain reaction (PCR) can be used to produce many copies of a gene for transfer.

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- (b) Some species of *Anopheles* mosquitoes, such as *Anopheles stephensi*, transmit malaria. Scientists have created genetically engineered *A. stephensi* mosquitoes in an attempt to reduce the spread of malaria.

A gene called *tTav* was constructed by scientists and transferred into the eggs of *A. stephensi*.

- *tTav* consists of DNA sequences from a bacterium, *Escherichia coli*, and herpes simplex virus, which is a DNA virus.
- The gene codes for tTav protein, which stops the expression of genes that are essential to mosquito development.
- Genetically engineered *A. stephensi* do not survive beyond the larval stage and therefore do not develop into adults.
- A chemical called tetracycline can stop the action of the tTav protein.

Male genetically engineered *A. stephensi* were released into the wild to breed with females. Their offspring had the *tTav* gene.

- (i) Describe how the *tTav* gene could have been synthesised and transferred into the eggs of *A. stephensi*.

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- (ii) Suggest how the tTav protein prevents the expression of other genes in *A. stephensi*.

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- (iii) Genetically engineered *A. stephensi* larvae were exposed to tetracycline in the laboratory.

Suggest why *A. stephensi* larvae were exposed to tetracycline in the laboratory.

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[Total: 11]